

Exercise 1: Market Pricing Strategy

A local manufacturing company in West Africa produces household cleaning products. The market is highly price-sensitive due to strong competition and limited purchasing power. The company currently sells its product at 1.50 USD per unit and sells approximately 5000 units per month. Market research shows that: for every 0.10 USD decrease in price, sales increase by about 1000 units, for every 0.10 USD increase, sales decrease by the same amount. The production cost per unit is 0.60 USD, and there is no production capacity limit. Decision problem Management wants to choose the selling price that maximizes monthly profit, not revenue.

Question

1. Translate the verbal market behavior into a mathematical demand model.
2. Derive the profit function.
3. Determine whether the problem is a maximization or minimization problem.
4. Analyze the shape of the objective function (convex/concave).
5. Find the optimal price analytically.
6. Verify the result numerically and graphically.

Exercise 2: Cybersecurity Alert Management (SOC Optimization)

A national bank in Africa operates a Security Operations Center (SOC). The SOC uses an automated system that assigns a risk score to each network event. Events with a score above a threshold are flagged for manual investigation. The SOC has: limited analysts, limited budget, strict regulatory obligations. From historical data: with a threshold of 0.6, the SOC detects 90% of attacks, but generates 300 alerts per day, increasing the threshold by 0.1 reduces alerts by 200 per day, but also reduces detected attacks by 15 per day. Each detected attack saves 500 USD in avoided losses, investigated alert costs 100 USD in analyst time.

Question:

1. What detection threshold should the SOC use to maximize net daily security benefit?
2. Identify the decision variable.
3. Build models for: number of attacks detected; number of alerts generated.
4. Construct a net benefit (objective) function.
5. Decide whether constraints are needed (capacity, minimum detection).
6. Solve the optimization problem to maintain at least 80 detected attacks/day
7. Interpret the result in operational terms.